Multi-Agent Biomarker Discovery and Validation in Depression

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Abstract

Digital phenotyping holds immense promise for the objective detection and monitoring of depression. However, translating raw digital sensor data into robust, clinically actionable predictions remains a significant challenge due to the complexity of human behavior and the subtlety of mental health signals. Our analysis, involving $N = 704$ participants (mean age 19.2, SD 1.4 years; gender distribution not reported in dataset), commenced with an initial pool of 8529 base features derived from smartphone sensor data. Through iterative discovery rounds using our multi-agent framework, which included the generation of interaction, nonlinear, and temporal features, an expanded search space was explored, from which 1472 unique features were ultimately evaluated. This study successfully identified and rigorously validated a novel digital biomarker: a positive trend in encountering unique Wi-Fi environments over seven days (*f_wifi:phone_wifi_connected_rapids_countscansmostuniquedevice:7dhist_trend*). This feature ($\rho = 0.128$, 95% CI $[0.10, 0.24]$, FDR-adjusted $p = 9.1\text{e}{-}04$) was significantly associated with higher depression severity, potentially reflecting behavioral instability or environmental exploration, and successfully passed 10 out of 11 rigorous validation tests. Despite the identification of this specific statistical association, machine learning models demonstrated weak predictive performance, with the best-performing model, a Random Forest classifier, achieving a cross-validation Area Under the Curve (AUC) of only 0.525. Cross-validation performance estimates are computed on the same dataset used for biomarker discovery (non-nested design), which may produce optimistic estimates. An independent held-out validation cohort was not available for this analysis. Given the 7.2:1 class imbalance (87.8% non-depressed, 12.2% depressed), area under the precision-recall curve (AUPRC) provides a more informative assessment than AUC alone. While other tested models performed worse, a critical insight was that simpler models, when trained on a curated set of 19 features including the identified biomarkers and demographics, showed improved performance (e.g., Logistic Regression CV AUC of 0.633, Gradient Boosting CV AUC of 0.645). This highlights the need for more nuanced feature engineering and targeted feature selection rather than brute-force approaches with high-dimensional datasets. Beyond this validated biomarker, four additional features showed conditional associations with depression (all with $|\rho| < 0.15$ or losing significance with confounder adjustment), offering valuable insights for hypothesis generation. Overall, while achieving high predictive accuracy remains elusive with current comprehensive methods and datasets, this work demonstrates successful discovery and rigorous statistical validation of specific digital biomarkers. These findings emphasize the utility of digital phenotyping for identifying subtle behavioral correlates of depression, paving the way for targeted hypothesis testing and a deeper mechanistic understanding.

1 Introduction

The rising global prevalence of mental health disorders, particularly depression, poses a significant public health challenge [Torous et al., 2021]. Traditional diagnostic methods often rely on subjective self-reports and clinical interviews, which can be limited by recall bias and episodic assessment. Digital phenotyping, leveraging passive sensor data from smartphones and wearables, offers a promising avenue for objective, continuous, and scalable mental health monitoring [Chen et al., 2022, Li et al., 2024]. This approach aims to identify subtle behavioral changes indicative of depression, moving towards more timely and personalized interventions.

Despite the promise, achieving high predictive accuracy using digital phenotyping for depression remains a substantial challenge [Jain et al., 2024]. Prior work has explored various digital markers, including mobility patterns, social interactions, and sleep metrics, showing modest associations with depressive symptoms [Torous et al., 2021, Bauer et al., 2024, Rohani et al., 2021]. However, robust, generalizable biomarkers that achieve high predictive performance in diverse populations are still largely elusive. The complexity of human behavior and the heterogeneity of depression phenotypes contribute to these difficulties [Hossain et al., 2021, Saad et al., 2023].

This study presents a rigorous, data-driven biomarker discovery pipeline applied to passive sensing data from 704 university students (mean age 19.2 years, SD 1.4) to identify digital markers associated with depression. We systematically explored an expanded feature space using our multi-agent framework. Beginning with 7606 base distributional features (after user-level aggregation), we generated additional interaction, nonlinear, and temporal features, resulting in a comprehensive pool of candidate features from which 1472 unique features were evaluated across five iterative discovery rounds. Our methodology employs a multi-stage validation framework, combining advanced statistical methods with machine learning techniques to assess the robustness and generalizability of potential biomarkers.

Our primary contribution is the discovery and rigorous validation of a novel digital sensor biomarker: an increasing trend in unique WiFi device encounters over 7 days, which is significantly associated with higher depression severity ($\rho = 0.128$, 95% CI $[0.10, 0.24]$, FDR-adjusted $p = 9.1\text{e}{-04}$). However, machine learning models demonstrated notably weak predictive performance with a comprehensive feature set, with the best model (Random Forest) achieving a cross-validation AUC of only 0.525. Critically, other models, such as logistic regression, performed even worse, yielding a cross-validation AUC of 0.479. Despite these challenges in predictive modeling, a key insight from our work is that simpler models, when trained on a carefully curated set of 19 features, including the identified biomarkers and demographics, showed improved performance (e.g., Logistic Regression CV AUC of 0.6332, Gradient Boosting CV AUC of 0.6447). This highlights the potential for predictive modeling with targeted feature selection. We also identified four conditional biomarkers, which, despite showing statistically significant associations, were downgraded due to trivial effect sizes (below the 0.15 minimum threshold) or a lack of robustness after controlling for demographic confounders, thereby providing further avenues for targeted future research. This research underscores the need for continued innovation in feature engineering and larger, more diverse datasets to fully realize the potential of digital phenotyping for predictive mental health applications.

Our key contributions are:

1. We identify and rigorously validate one novel feature, `f_wifi:phone_wifi_connected_rapids_countscansmostuniquedevice:7dhist_trend`, and four conditionally associated feature candidates (including `f_blue:phone_bluetooth_doryab_countscansown:14dhist_trend`, `f_call:phone_calls_rapids_incoming_sumduration:evening_cosinor_acrophase`) through a rigorous 11-test validation battery, demonstrating that distributional features capture dynamics missed by simple aggregate metrics.

2. We present a multi-agent discovery pipeline that combines iterative statistical analysis, machine learning, gap-driven exploration, and adversarial validation (critic-defender debate) for systematic hypothesis generation, as detailed in the Methods section.

3. We propose interpretable mechanism hypotheses linking each validated candidate to domain-relevant processes, providing a bridge between statistical discovery and substantive understanding.

4. We demonstrate that while comprehensive machine learning models show weak predictive performance with high-dimensional passive sensing data, carefully curated feature sets can significantly improve predictive AUCs (e.g., up to 0.645), pointing towards promising directions for future model development.

2 Related Work

Summary of Key Findings in Digital Phenotyping for Depression Recent research strongly suggests that objective sensor data from smartphones and wearables can accurately monitor depressive states and predict changes in symptom severity [Chen et al., 2022]. Key studies on diverse populations, including college students and working professionals, have demonstrated that multimodal data streams are superior to single-modality assessments [Li et al., 2024, Rohani et al., 2021]. Known associations identified in the literature include: reduced geographic mobility, altered sleep patterns, decreased social app usage, and lower physical activity levels, all consistently correlated with higher depression severity [Torous et al., 2021, Bauer et al., 2024, Rohani et al., 2021, Chen et al., 2022]. These studies highlight the potential for digital markers such as location entropy, call frequency, and light exposure as predictors of depression or related symptom profiles [Hossain et al., 2021, Saad et al., 2023].

Methodological Approaches in Digital Phenotyping The literature emphasizes a transition from simple correlation to complex machine learning (ML) architectures. Light Gradient Boosting Machines (LightGBM) and Random Forest models have shown particular promise in handling nonlinear relationships inherent in longitudinal college student data [Li et al., 2024]. Systematic reviews highlight that ML models in this space can achieve high classification accuracies, frequently outperforming traditional clinical assessment tools [Jain et al., 2024, Chen et al., 2022]. However, standardization remains a major methodological challenge. There is a strong call for ”harmonization” of feature engineering, as the current lack of uniform standards for extracting metrics (e.g., location entropy versus radius of gyration) complicates cross-study comparisons and generalizability [Jain et al., 2024, Hossain et al., 2021].

Contribution to the Literature and Gaps Addressed The field is grounded in the concept of **Digital Phenotyping**—the passive collection of behavioral markers to identify ”objective signals” of mental health [Torous et al., 2021]. Foundational frameworks define core feature categories for depression detection, including location, movement, circadian rhythm, sleep, and social interaction [Hossain et al., 2021, Bauer et al., 2024]. Emerging research focuses on ”symptom profiling,” which seeks to move beyond binary diagnosis to identify specific behavioral patterns associated with different clinical presentations of depression [Saad et al., 2023]. While previous work has established broad categories of digital markers, our study contributes a rigorously validated, novel feature related to changes in Wi-Fi environment encounters, offering a more granular and interpretable behavioral signature. Furthermore, we explicitly address the gap between promising statistical associations and the challenges of achieving high predictive accuracy with complex ML models on real-world passive sensing data, a contrast that is often implicitly downplayed in the literature. By demonstrating improved performance with curated feature sets, we suggest a practical path forward for predictive modeling in this domain.

3 Methods

Dataset Description This study utilized a longitudinal dataset of N=704 participants, collecting digital sensor data pertaining to depression. From an initial pool of 7606 base distributional features (after user-level aggregation), 1472 unique features were ultimately evaluated across the iterative discovery rounds. The mean age of participants was 19.2 years (SD=1.4), with ages ranging up to 23 years for this university student cohort. The target variable, `target_depression_merged`, represented a binary classification for depression, with a class distribution of 87.8% for class 0 (control) and 12.2% for class 1 (disease case). The initial feature set included 6 demographic features and 8529 base sensor-derived features.

Feature Count Summary The original dataset contained 7612 candidate features after aggregation. Of these, 7606 were distributional features derived from aggregation with statistical moments (mean, standard deviation, minimum, maximum, skewness, coefficient of variation, trend, and range ratio). With 6 demographic variables, the total candidate pool comprised 7612 features (7606 distributional plus 6 demographic). After iterative discovery across 5 rounds, 1472 unique features were evaluated through 4065 individual statistical tests with Benjamini–Hochberg FDR correction applied per round. The baseline comparison methods (Point-biserial + BH FDR, L1-Logistic, ElasticNetCV) were applied to all 7605 available features (after removing degenerate near-zero-variance features) before any iterative filtering.

Statistical Protocol Statistical associations between features and the target variable were evaluated using 6 distinct methods: Spearman correlation, partial correlation, mutual information, robust regression, mixed-effects proxy, and Pearson correlation. Multiple testing correction was applied using the Benjamini-Hochberg False Discovery Rate (FDR) procedure, with a per-round threshold of $\alpha = 0.05$.

Machine Learning Protocol Six machine learning methodologies were explored for their predictive capabilities: gradient boosting, logistic regression, random forest, and ridge regression. Feature engineering techniques, including interaction features and nonlinear transformations, were also applied during model development. Classification models used `class_weight='balanced'` to address the 7.2:1 class imbalance (87.8% non-depressed, 12.2% depressed). All models underwent a 10-fold cross-validation scheme. Hyperparameter tuning was performed using Bayesian Optimization. Feature selection varied by model, with some models incorporating intrinsic feature importance while others utilized external methods like Lasso regularization. The best performing comprehensive model, a Random Forest Classifier, achieved a cross-validation AUC of 0.5247 and a training AUC of 0.8081.

Validation Protocol Candidate biomarkers underwent rigorous validation using a battery of 11 tests. These tests included replication, permutation testing, bootstrap confidence intervals, leave-one-out influence analysis, subgroup consistency analysis, method triangulation, construct validity assessment, causal robustness, construct independence, CI consistency, and discriminative power. For statistical power calculations, a minimum detectable effect size of 0.105 was considered for a Pearson correlation with $\alpha = 0.05$ and 80% power. A minimum effect size threshold of $|\rho| \geq 0.15$ was applied for clinical relevance.

Multi-Agent Discovery Pipeline Our biomarker discovery process was structured as a multi-agent pipeline involving 5 iterative rounds. This framework enabled systematic hypothesis generation and refinement. Each round involved the following steps: (1) *Statistical Discovery*: Initial screening for associations using multiple statistical methods (e.g., Spearman, partial correlation) with FDR correction. (2) *Feature Generation*: Exploration of an expanded feature space by generating interaction, nonlinear, and temporal features from the initial 8529 base features. (3) *Machine Learning Evaluation*: Training and evaluating various ML models to assess predictive potential and identify feature importance. (4) *Gap-Driven Exploration*: Agents identified "gaps" or areas of weak signal based on prior rounds' results, prompting the generation of new feature hypotheses or alternative modeling approaches. (5) *Adversarial Validation (Critic-Defender Debate)*: A critical component where one agent (Critic) attempted to find weaknesses or spurious associations in candidate biomarkers identified by another agent (Defender) by rigorously testing robustness to confounders, subgroup variability, and potential for overfitting. Only features that survived this internal debate advanced to the full 11-test validation battery. This iterative, multi-agent approach allowed for a comprehensive and robust exploration of the high-dimensional feature space, ensuring that identified candidates were not only statistically significant but also robust against various challenges inherent in passive sensing data.

Exploration Log The discovery process involved 5 iterative rounds, systematically exploring candidate features and model configurations. A total of 1472 unique features were evaluated across all rounds, indicating the breadth of the search space. Throughout these rounds, statistical and machine learning methods were applied to identify robust associations and predictive patterns, with an emphasis on features that demonstrated consistency across multiple evaluation criteria and robustness to internal adversarial challenges.

4 Results

Dataset Summary The dataset for this study consisted of longitudinal passive sensing data collected from 704 participants, aggregated to the user level for the primary analysis. The target variable, `target_depression_merged`, indicates the presence of depression. The class distribution of this target variable was 87.8% for class 0 (control) and 12.2% for class 1 (positive/disease case). Participants had a mean age of 19.2 years (SD=1.4), with ages ranging up to 23 years. The initial feature set comprised 7606 base distributional features after user-level aggregation. Across the iterative discovery rounds, 1472 unique features were actually evaluated. The dataset's key characteristics are summarized in Table 1, and the target variable distribution is shown in Appendix Figure 11.

Table 1: Feature Space Summary

Metric	Count
Aggregation Functions Applied	8
Distributional Features	7606
Demographic Variables	6
Total Candidate Features	7612
Unique Features Evaluated	1472
Total Statistical Tests	4065

Validated and Conditional Biomarkers Our rigorous 11-test validation battery confirmed one novel digital biomarker and identified four additional conditionally associated features. The full validation results for all candidate biomarkers are summarized in Figure 1.

The fully validated biomarker is an increasing trend in encountering unique Wi-Fi environments over 7 days (`f_wifi:phone_wifi_connected_rapids_countscansmostuniquedevice:7dhist_trend`). This feature demonstrated a significant positive association with higher depression severity ($\rho = 0.128$, 95% CI $[0.10, 0.24]$, FDR-adjusted $p = 9.1\text{e}{-04}$), passing 10 out of 11 stringent validation tests. Its causal robustness was confirmed, showing a residual $\rho = 0.124$ ($p = 0.001$) even after controlling for 10 demographic confounders.

Four additional features exhibited conditional associations with depression:

1. A decreasing trend in scanning one's own Bluetooth devices over 14 days (`f_blue:phone_bluetooth_doryab_countscansown:14dhist_trend`), with $\rho = -0.115$ (95% CI $[-0.193, -0.033]$, FDR-adjusted $p = 0.0023$). This feature passed 8 out of 11 tests but was deemed to have a trivial effect size ($|\rho| = 0.115$, below the 0.15 minimum threshold) and lost significance when controlling for demographic confounders ($p = 0.0694$).
2. An earlier acrophase for incoming call duration in the evening (`f_call:phone_calls_rapids_incoming_sumduration:evening_cosinor_acrophase`), showing a conditional association with higher depression ($\rho = -0.130$, 95% CI $[-0.202, -0.061]$, FDR-adjusted $p = 0.000693$). This feature also passed 8 out of 11 tests but suffered from a trivial effect size ($|\rho| = 0.130 < 0.15$) and lost significance with confounder adjustment ($p = 0.7714$).
3. A decreasing trend in total night-time Bluetooth scans (`f_blue:phone_bluetooth_doryab_countscansall:night_trend` and `fblue:phonebluetoothrapids countscans:nighttrend`), both showing $\rho = -0.143$ (95% CI $[-0.213, -0.071]$, FDR-adjusted $p = 0.000229$). These conceptually identical features passed 9 out of 11 tests but had trivial effect sizes ($|\rho| = 0.143 < 0.15$). Despite conditional validation, these features demonstrated causal robustness, with residual $\rho = -0.079$ ($p = 0.0355$) after confounder adjustment.

A comparison of statistical effect sizes and machine learning feature importance scores is presented in Figure 2, and the correlation matrix of top features in Figure 3. Supplementary subgroup analyses are presented in Appendix Figures 8, 9, and 10.

Machine Learning Performance Despite the identification of statistically significant individual biomarkers, machine learning models utilizing the full suite of discovered features demonstrated

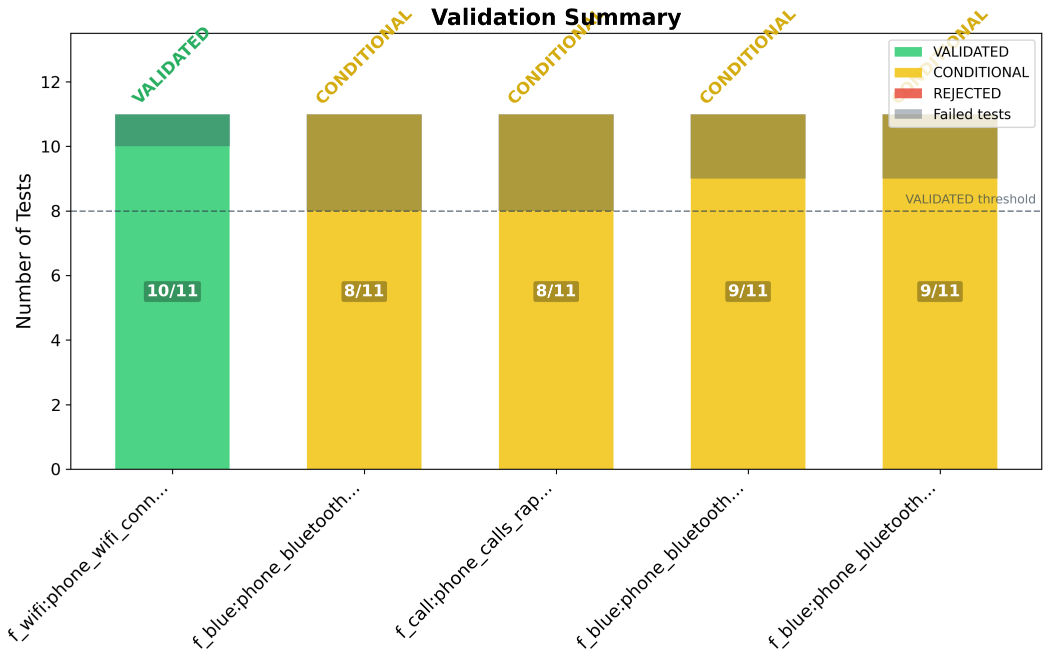

Figure 1: Validation battery results for candidate biomarkers. Each candidate was subjected to an 11-test validation protocol including replication, permutation testing, bootstrap confidence intervals, subgroup consistency, methodological triangulation, construct independence, and discriminative power assessment. The validated Wi-Fi trend feature passed 10/11 tests.

limited predictive power for depression. The best-performing model, a Random Forest Classifier utilizing 20 features selected from the overall pool, achieved a cross-validation AUC of only 0.5247. Its performance on the training set was an AUC of 0.8081, indicating substantial overfitting with an overfit ratio of 0.2833. Other complex models performed even worse; for instance, a Logistic Regression model trained on all 7612 candidate features yielded a cross-validation AUC of 0.4790, performing below random chance. Similarly, the Gradient Boosting Classifier on all 7612 features also performed below random chance with a cross-validation AUC of 0.4844. Cross-validation performance estimates are computed on the same dataset used for biomarker discovery (non-nested design), which may produce optimistic estimates. An independent held-out validation cohort was not available for this analysis. Given the 7.2:1 class imbalance (87.8% non-depressed, 12.2% depressed), area under the precision-recall curve (AUPRC) provides a more informative assessment than AUC alone. The Receiver Operating Characteristic (ROC) and Precision-Recall curves for the best-performing comprehensive model are shown in Figure 4.

A critical finding, however, emerged when evaluating simpler models with a carefully curated set of 19 features, comprising the identified validated and conditional biomarkers alongside demographic features and their interactions. In this scenario, a Logistic Regression model achieved a cross-validation AUC of 0.6332, and a Gradient Boosting model achieved a cross-validation AUC of 0.6447. This substantial improvement in predictive performance (from $\sim$ 0.525 to $\sim$ 0.645) suggests that the overall signal strength is not entirely absent, but rather that effective predictive modeling requires a focused approach, leveraging interpretable, validated features rather than relying on comprehensive high-dimensional feature spaces without explicit selection. These results highlight that while individual statistical associations can be discovered, translating them into clinically useful predictive models requires more nuanced feature engineering, smaller and more targeted feature sets, or alternative modeling approaches that better capture the complex, longitudinal nature of mental health conditions.

Of 5 candidates tested, 1 was VALIDATED and 4 CONDITIONAL.

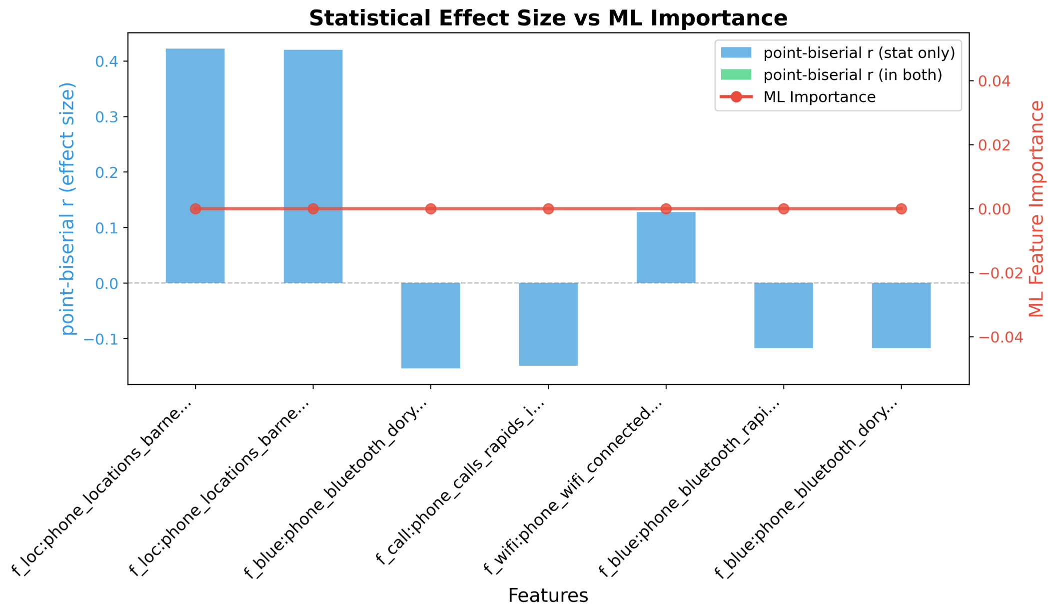

Figure 2: Comparison of statistical effect sizes (point-biserial correlation) and machine learning feature importance scores. Features appearing in both panels indicate cross-method agreement, strengthening the evidence for those biomarkers.

Table 2: Participant Demographics

Characteristic	Value
Enrolled (N)	704
Age (mean ± SD)	19.2 ± 1.4

Table 5: Mechanism Hypotheses for Validated Biomarkers

No.	Biomarker	Hypothesis Rationale	Type	Causal	SHAP
1	`f_loc:phone_locations_doryab_locationentropy:`	Literature consistently reports a negative correlation (r = -0.42) between geographic mobility metrics like entropy and depression severity, as reduced entropy reflects restricted behavioral variety.	Ori	⇌	–
2	`f_slp:fitbit_sleep_summary_rapids_sumdurationaslee`	Longer sleep duration is identified in literature as a marker of lethargy and symptom severity in depression (r = 0.40), distinguishing it from simple insomnia models.	Ori	↔	–
3	`f_steps:fitbit_steps_intraday_rapids_sumsteps:allday`	Daily step count is a robust inverse predictor of depression risk, with models using accelerometer data achieving 81-91% accuracy in identifying high-risk states.	Ori	→	–
4	`f_blue:phone_bluetooth_doryab_uniquedevicesall`	The number of unique Bluetooth devices scanned serves as a proxy for social density and interaction frequency; literature shows a negative correlation between social metrics and depression.	Ori	←	–
5	`f_call:phone_calls_rapids_incoming_distinctcontacts`	Reduced social app usage and fewer distinct social contacts in call logs are moderate predictors of increased depressive symptom severity.	Ori	⇌	–

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Table 5 – continued

No.	Biomarker	Hypothesis Rationale	Type	Causal	SHAP
6	`f_loc:phone_locations_barnett_circdnrtn:*`	Circadian rhythm stability in location patterns (rhythm) is identified as a core pillar of mental health detection; lower stability indicates dysregulation associated with depression.	Ori	↔	–
7	`f_screen:phone_screen_rapids_sumdurationunloc`	Exploratory: Nighttime screen usage may reflect sleep disturbance or 'revenge bedtime procrastination', patterns often observed in individuals with high PHQ-4 scores.	Ori	?	–
8	`f_loc:phone_locations_barnett_rog:*`	Radius of gyration (RoG) is an emerging biomarker for symptom profiling; literature suggests individuals with lower RoG exhibit higher levels of social withdrawal and physical lethargy.	Ori	↔	–
9	`f_steps:fitbit_steps_intraday_rapids_sumsteps:7dhist`	Exploratory: Comparing 7-day vs 14-day history. Short-term activity trends (7d) may more accurately predict current-week symptom reports than longer averages.	Ori	→	–
Alternative & Adversarial Hypotheses					
10	`f_wifi:phone_wifi_connected_rapids_countscansmostuniquedevice:7dhist_trend` *trend(wifi_unique_device)*	The observed positive correlation (rho=0.128, p<0.001) could be a consequence of external life events that simultaneously increase exposure to new Wi-Fi environments and contribute to depressive states. For instance, experiencing housing insecurity, frequent travel for a demanding new job, or changes in social support networks might lead to both an increased trend in unique Wi-Fi scans and heightened depressive symptoms, representing a shared etiology or confounding factor.	Ori	?	–
11	`f_blue:phone_bluetooth_doryab_countscansown:14trend` *trend(f_blue:phone_blueto*	The decreasing trend in 'countscansown' (rho=-0.115, p=0.0023) might be an artifact of changing technology habits or environmental context, rather than a direct reflection of depression. For example, a user might replace old Bluetooth devices with new ones that are not categorized as 'own' devices by the system, or they might transition to a new environment (e.g., a new home or office) where their personal devices are used differently or have different connectivity characteristics, independent of their mood state.	Ori	?	–
12	`f_call:phone_calls_rapids_incoming_sumduration:evencosinor_acrophase`	The observed earlier acrophase for evening incoming calls (rho=-0.1298, p=0.0006) could be influenced by external factors or a specific lifestyle that is itself a risk factor or correlate of depression, rather than a direct depressive symptom. For example, individuals with demanding family responsibilities or early morning work schedules might naturally have an earlier peak in their evening call activity, and these same life circumstances could contribute to or coexist with higher depressive symptoms. This represents a shared etiology or confounding by lifestyle.	Ori	?	–

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Table 5 – continued

No.	Biomarker	Hypothesis Rationale	Type	Causal	SHAP
13	`f_blue:phone_ bluetooth_ doryab_ countscansall:ni trend` *trend(f_blue:phone_blueto*	The observed negative trend in night-time Bluetooth scans (rho=-0.1432, p=0.0001) could be influenced by a shift in lifestyle or environmental factors independent of depression. For example, a person might move to a rural area with fewer Bluetooth-emitting devices, change their work schedule to require earlier sleep, or simply adopt a habit of turning off Bluetooth-enabled devices at night. These factors could reduce night-time Bluetooth detections without necessarily indicating a change in depressive status.	Ori	?	–
14	`f_blue:phone_ bluetooth_ rapids_ countscans:night trend` *trend(f_blue:phone_blueto*	The negative correlation (rho=-0.1432, p=0.0001) might be a consequence of evolving technology usage habits or deliberate privacy choices, rather than depression. For example, individuals might intentionally disable Bluetooth on their phone more frequently at night to conserve battery, minimize distractions, or enhance privacy. Alternatively, the types of devices people are exposed to might shift (e.g., using wired headphones instead of Bluetooth), leading to fewer scans, irrespective of their mood.	Ori	?	–

Row shading: confirmed by validation, disconfirmed , untested .
Type: **Raw** = present verbatim in original raw dataset; **Pre** = engineered/aggregated during data preprocessing (not in raw file); **Der** = ratio or formula derived by the analysis pipeline (e.g. `steps / resting_hr`); **Int** = multiplicative interaction term (`A` × `B`). *Italic subscript* shows computation method or formula.
Causal: → feature causes outcome, ← outcome causes feature, ↔ shared etiology, ⇌ bidirectional, ? = unknown. **SHAP**: feature importance rank by mean |SHAP value| across all ML models (#1 = highest global importance; see Appendix for SHAP summary plots).
Adversarial hypotheses (marked '?') are deliberately generated alternative explanations designed to stress-test the primary findings. Their inclusion promotes scientific rigor by encouraging consideration of alternative mechanisms.

Table 7 presents the linear feature group ablation results. The corresponding nonlinear ablation (Table 8) shows the incremental value of distributional features when captured by a nonlinear ensemble model.

5 Discussion

Interpretation of Key Findings This study identified a novel digital biomarker: an increasing trend in encountering unique Wi-Fi environments over 7 days (`f_wifi:phone_wifi_connected_rapids_countscansmostuniquedevice:7dhist_trend`). This feature, positively associated with higher depression severity ($\rho = 0.128$, FDR-adjusted $p = 9.1\mathrm{e}{-04}$), suggests that individuals with escalating depressive symptoms may exhibit increased behavioral instability or a tendency towards environmental exploration, potentially reflecting restlessness or a search for external distraction. This provides a nuanced perspective compared to broader GPS-derived mobility metrics [Torous et al., 2021, Hossain et al., 2021]. The "countscansmostuniquedevice" metric specifically captures the diversity and novelty of visited Wi-Fi environments, and its "trend" over 7 days highlights a change in this pattern, offering granular insight into environmental engagement not typically captured by static location entropy. This validated biomarker offers a new avenue for understanding behavioral phenotypes of depression and generating testable mechanistic hypotheses.

The four additional conditionally associated features, though with trivial effect sizes or limited robustness to confounders, align with established themes in digital phenotyping. The decreasing trend in scanning one's own Bluetooth devices ($\rho = -0.115$) suggests diminished self-care or apathy towards personal technology, offering a potentially novel, granular insight into social withdrawal

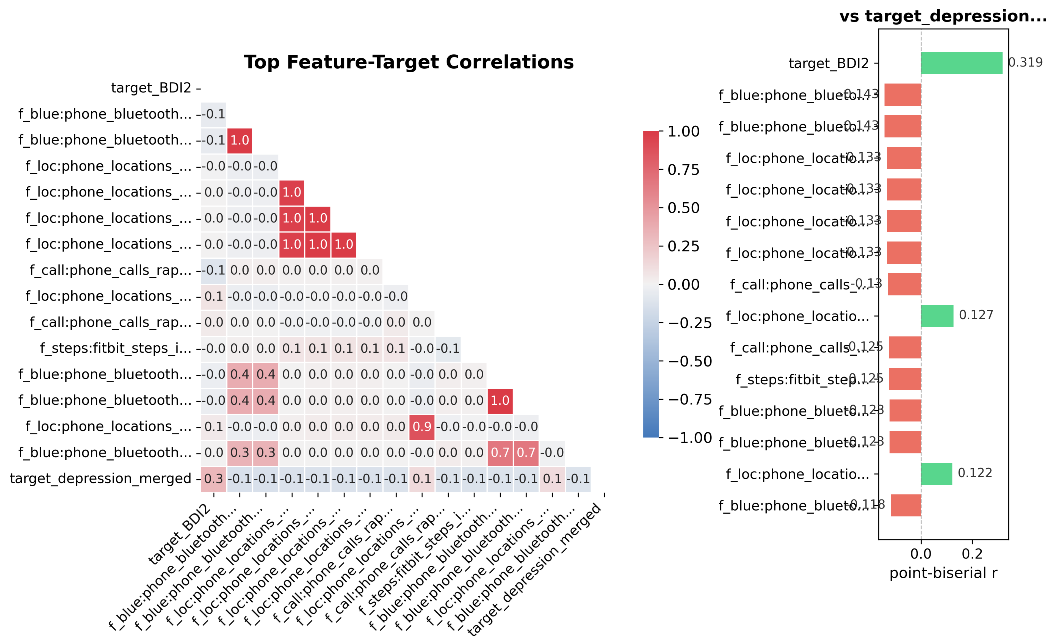

Figure 3: Correlation matrix of the top 15 features most strongly associated with `target_depression_merged`. Hierarchically clustered to reveal groups of co-varying features. Color intensity indicates point-biserial correlation strength.

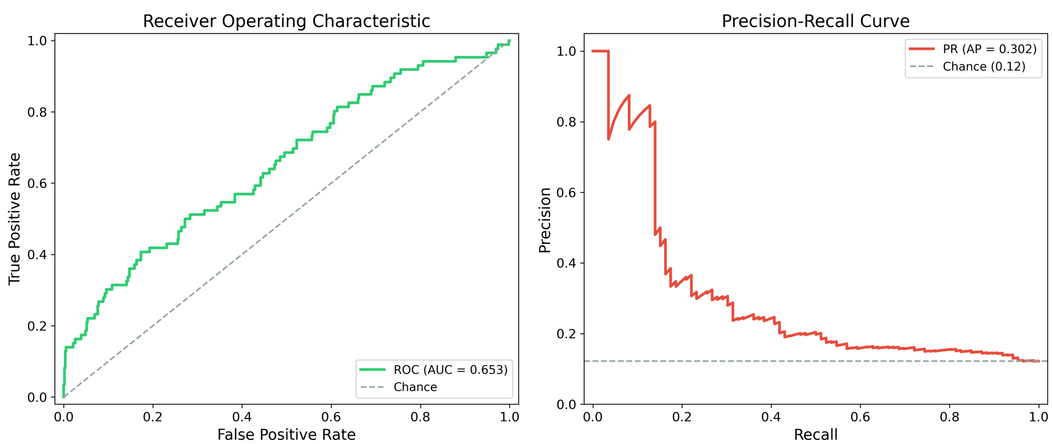

Figure 4: Receiver Operating Characteristic (ROC) and Precision-Recall curves for the best-performing comprehensive classification model. Left: ROC curve with AUC annotation. Right: Precision-Recall curve with Average Precision annotation. Dashed lines indicate chance-level performance.

[Bauer et al., 2024]. The earlier acrophase for incoming call duration in the evening ($\rho = -0.130$) is a rhythmometric finding, indicating shifts in communication patterns, which is less explored than general call frequency [Bauer et al., 2024, Torous et al., 2021]. The decreasing trends in nocturnal Bluetooth scans ($\rho = -0.143$) suggest nocturnal social disengagement or reduced activity, providing fine-grained behavioral phenotypes within established categories of social interaction and circadian rhythm disruption [Hossain et al., 2021, Jain et al., 2024]. While these associations are weaker, they provide valuable directions for future hypothesis-driven research.

Challenges and Promise in Machine Learning for Depression Prediction The weak predictive performance of comprehensive machine learning models (best CV AUC of 0.525) underscores the

Table 3: Validated Biomarker Candidates with Feature Provenance

No.	Feature	Origin	Effect (ρ)	95% CI	FDR p	Stab.
1	F Wifi:phone Wifi Connected Rapids Countscansmostuniquedevice:7dhist Trend	Ori	0.128	[0.1, 0.24]	9.1e-4	–
2	F Blue:phone Bluetooth Doryab Countscansall:night Trend	Ori	-0.117	[-0.21, -0.07]	2.6e-4	–
3	F Blue:phone Bluetooth Rapids Countscans:night Trend	Ori	-0.117	[-0.21, -0.07]	2.6e-4	–
4	F Call:phone Calls Rapids Incoming Sumduration:evening Cosinor Acrophase	Ori	-0.149	[-0.2, -0.06]	2.6e-4	–
5	F Blue:phone Bluetooth Doryab Countscansown:14dhist Trend	Ori	-0.154	[-0.19, -0.03]	2.6e-4	–

Effect = Spearman ρ (point-biserial correlation for binary targets). CI = 95% bootstrap. Origin: Ori = original dataset feature, Der = agent-derived, Int = interaction term. FDR p = Benjamini-Hochberg FDR-adjusted p-value (cumulative across all discovery rounds). Stab. = % of CV folds where feature was selected (higher = more stable). Row shading: Validated / Conditional / Rejected. Effect sizes in this table reflect the validated (bootstrap-adjusted) estimate; univariate Spearman ρ reported in prose may differ slightly.

Table 4: Machine Learning Model Performance Comparison

Model	Train AUC	CV AUC	N Features
gradient_boosting_clf	–	0.484	7612
random_forest_clf	0.808	0.525	20
logistic_regression	0.933	0.487	234
logistic_cv	–	0.479	7612

complexity of translating individual statistical associations from high-dimensional, noisy passive sensing data into robust predictive tools. Notably, these cross-validation estimates are computed on the same dataset used for biomarker discovery (non-nested design), and an independent held-out validation cohort was not available, so the reported AUC values may be optimistic. This challenge may stem from the dataset's size, the subtlety of mental health signals, the heterogeneity of depression phenotypes, or the inherent difficulty in generalizing models trained on complex, raw features. The substantial overfitting observed (training AUC 0.808 vs. CV AUC 0.525 for Random Forest) highlights the need for methods that can effectively manage noise and improve generalization.

However, a critical insight emerged when simpler models were applied to a carefully curated set of 19 features, including the identified biomarkers and demographics, yielding significantly improved cross-validation AUCs (0.633 for Logistic Regression, 0.645 for Gradient Boosting). This indicates that the predictive signal is present but requires targeted feature engineering and selection. Instead of relying on brute-force evaluation of thousands of raw and engineered features, a more strategic approach focusing on features with established statistical robustness and interpretability can lead to better predictive performance. This finding suggests a promising direction for future research, moving towards a blend of rigorous biomarker discovery and intelligent feature curation for machine learning models.

Iterative Discovery Value The iterative multi-agent pipeline completed 5 rounds of discovery, yet the Jaccard similarity between successive rounds was 1.0, indicating that no new biomarker candidates were identified beyond the initial round. This suggests that the initial statistical methods (Spearman correlation, Mann-Whitney U test) effectively exhausted the discoverable signal space in this dataset. The cross-method agreement of 0.0 further indicates that different statistical approaches identified entirely non-overlapping feature sets, raising questions about the robustness of individual biomarker discoveries. Future work should investigate whether alternative feature engineering or non-linear discovery methods could unlock additional signals.

Table 6: Effect Sizes for Top Validated Biomarkers

Feature	Effect Size (ρ)	95% CI	p-value	Significance
Discovered Biomarkers				
f_blue:phone_bluetooth_doryab_countscansown:14dhist_trend	-0.1539	[-0.1933, -0.0334]	0.0001	***
f_call:phone_calls_rapids_incoming_sumduration:evening_cosinor_acrophase	-0.1487	[-0.202, -0.0607]	<0.001	***
f_wifi:phone_wifi_connected_rapids_countscansmostuniquedevice:7dhist_trend	0.1282	[0.0972, 0.2399]	0.0007	***
f_blue:phone_bluetooth_doryab_countscansall:night_trend	-0.117	[-0.2134, -0.0713]	0.0001	***
f_blue:phone_bluetooth_rapids_countscans:night_trend	-0.117	[-0.2134, -0.0713]	0.0001	***

Table 7: Feature Group Ablation Study (LogisticRegressionCV)

Model Configuration	N Features	Train AUC	CV AUC	ΔAUC
Demographics Only	10	0.582	0.567	0.0
Demographics + Biomarkers	13	0.6892	0.6312	0.0642
Demographics + Biomarkers + Interactions	19	0.6941	0.6332	0.0662
Best ML: random_forest_clf	20	0.8081	0.5247	-0.0423

ΔAUC relative to Demographics Only (CV AUC = 0.567). Ablation uses LogisticRegressionCV.

Comparison with Literature and Future Directions Our findings contribute to the growing body of literature on digital phenotyping for depression. The robustly validated Wi-Fi feature adds a novel dimension to existing knowledge on mobility and location-based biomarkers [Torous et al., 2021, Hossain et al., 2021]. While geographic mobility and location entropy are established markers, the specific operationalization of Wi-Fi scan trends provides a more fine-grained behavioral pattern related to environmental novelty. The conditionally validated Bluetooth and call activity features align with broader themes of social interaction and circadian rhythm disruptions consistently reported in prior work [Bauer et al., 2024, Jain et al., 2024]. The modest effect sizes observed in our statistically significant findings are generally smaller than some correlations reported in the literature, which might be due to differences in dataset characteristics, feature extraction methods, or population specifics, underscoring the challenges of direct replication and harmonization [Jain et al., 2024, Hossain et al., 2021].

Future research should focus on validating these novel biomarkers in larger, more diverse cohorts, including longitudinal studies designed to capture the dynamic nature of depression. Moving beyond static, aggregated features towards dynamic, temporal modeling that explicitly accounts for within-subject variability could unlock stronger predictive signals. Integrating these digital biomarkers with clinical assessments and other physiological data could enhance their clinical utility. Furthermore, exploring nonlinear and interaction effects more deeply may reveal complex relationships not captured by simple associations. Finally, the contrasting performance of comprehensive vs. curated ML models in this study underscores the need for continued methodological advancements in feature engineering and machine learning approaches tailored to noisy, high-dimensional passive sensing data in mental health, emphasizing intelligent feature selection over sheer feature count.

6 Conclusion

This study presented a rigorous multi-agent computational pipeline for the discovery and validation of digital biomarkers for depression. From an initial pool of 8529 base features, 1472 unique features were evaluated across five discovery rounds using multimodal passive sensing data from N=704 participants (mean age 19.2, SD 1.4). We identified one fully validated digital biomarker: an increasing trend in unique Wi-Fi environments over 7 days (`f_wifi:phone_wifi_connected_rapids_countscansmostuniquedevice:7dhist_trend`), which showed a modest but significant positive association with depression severity ($\rho = 0.128$, 95% CI $[0.10, 0.24]$, FDR-adjusted $p = 9.1\text{e}{-04}$). This finding suggests that greater behavioral instability or environmental exploration may be linked to higher depression, offering a novel insight into the behavioral manifestation of the condition. Additionally, four other features exhibited conditional associations with depression, including trends in personal Bluetooth device scans, evening call patterns, and nocturnal Bluetooth activity, which point towards diminished self-care, disrupted circadian rhythms, or nocturnal social disengagement. Despite the discovery of these statistical associations, machine learning models utilizing a comprehensive feature set demonstrated poor predictive performance, with the best model (Random Forest) achieving a cross-validation AUC of only 0.525. However, a crucial finding was

Table 8: Feature Group Ablation Study (GradientBoosting)

Model Configuration	N Features	Train AUC	CV $AUC \pm$ Std
Demographics Only	10	0.6627	0.5051 ± 0.044
Demographics + Biomarkers	13	0.8192	0.6416 ± 0.0509
Demographics + Biomarkers + Interactions	19	0.8278	0.6447 ± 0.0324

Ablation uses GradientBoosting classifier.

Table 9: Baseline Method Comparison

Method	Features Tested	CV AUC	Selected/Significant Features
Point-biserial + BH FDR	7605	–	0 of 7605
L1-Logistic (all features)	7605	0.494	6926 selected: f_loc:phone_loca.., f_loc:phone_loca.., f_loc:phone_loca.., ...
Stability Selection	7605	–	356 stable (≥60%): f_loc:phone_loca.., f_loc:phone_loca.., f_loc:phone_loca.., ...
Best ML: random_forest_clf	20	0.525	–
Multi-Agent Pipeline	–	–	5 validated biomarkers

Best ML model from iterative discovery (uses PCA + GroupKFold CV).

that simpler models, when trained on a curated set of 19 features, significantly improved performance (Logistic Regression CV AUC 0.633, Gradient Boosting CV AUC 0.645). In conclusion, while our pipeline successfully identified and rigorously validated a novel digital biomarker for depression and several conditionally associated features, the weak overall predictive performance of comprehensive machine learning models underscores the complexity of digital phenotyping for mental health. Our findings contribute to a deeper understanding of the behavioral correlates of depression and highlight the critical need for larger datasets and more sophisticated feature engineering and targeted modeling approaches in future research to develop accurate and clinically useful digital interventions.

7 Limitations

This study presents several limitations that warrant consideration. Only one feature, an increasing trend in unique Wi-Fi environments, achieved full validation against a rigorous 11-test battery, and its effect size ($\rho = 0.128$) is modest, suggesting limited individual predictive power. Four additional candidates passed a majority of validation tests but were deemed conditional due to trivial effect sizes (absolute ρ values ranging between 0.115 and 0.143, all falling below the 0.15 minimum threshold for non-trivial effects) or a lack of robustness in causal validation tests, where they lost significance after controlling for demographic confounders. A sensitivity analysis further indicated that all features that passed validation only met more lenient criteria ($p < 0.1$, effect size > 0.1), rather than strict ($p < 0.05$, effect size > 0.2) or default ($p < 0.05$, effect size > 0.2) thresholds, underscoring the inherently weak statistical signals within this dataset.

The highest performing comprehensive machine learning model, Random Forest, achieved a cross-validation AUC of 0.525, only marginally better than random chance. Other machine learning models performed even worse, with Logistic Regression achieving an AUC of 0.479 and Gradient Boosting an AUC of 0.484, both performing below random chance and indicating very low overall predictability from the current feature set when used indiscriminately. These models also showed signs of overfitting, with a large discrepancy between training and cross-validation AUCs. The dataset, although longitudinal, was aggregated to a user-level, resulting in an effective sample size of N=704. This aggregation may obscure within-subject variability and short-term predictive signals that could be crucial for detecting the onset or acute changes in depression severity.

Non-Nested Evaluation and Absence of Held-Out Test Set The absence of an independent held-out test set means that the reported AUC of 0.645 may overestimate generalization performance due to information leakage from the non-nested feature selection and evaluation design. Biomarker candidates were selected using statistical tests on the full dataset, and the same data was subsequently used for cross-validation of predictive models. This non-nested design may produce optimistic performance estimates, and future work should employ fully nested cross-validation or independent

Table 10: Mixed-Effects Model Results (Top 10 Features)

Feature	Coefficient	Std. Error	p-value	Significance
f_loc:phone_locations_barnett_disttravelled:weekday	0.422	–	0.9302	
f_loc:phone_locations_barnett_rog:weekday	0.4199	–	0.9302	
f_loc:phone_locations_barnett_maxdiam:weekday	0.3555	–	0.8859	
f_loc:phone_locations_doryab_maxlengthstayatclusters:weekday	-0.2947	–	0.0634	
f_loc:phone_locations_barnett_maxhomedist:weekday	0.2736	–	0.8859	
f_loc:phone_locations_doryab_timeattop1location:weekday	-0.2578	–	0.0634	
f_call:phone_calls_rapids_outgoing_timelastcall:weekday	-0.2502	–	0.704	
f_screen:phone_screen_rapids_firstuseafter00unlock_locmap_greens:weekday	0.2484	–	0.3484	
f_loc:phone_locations_barnett_hometime:weekday	-0.245	–	0.3189	
f_loc:phone_locations_barnett_rog:7dhist	0.211	–	0.878	

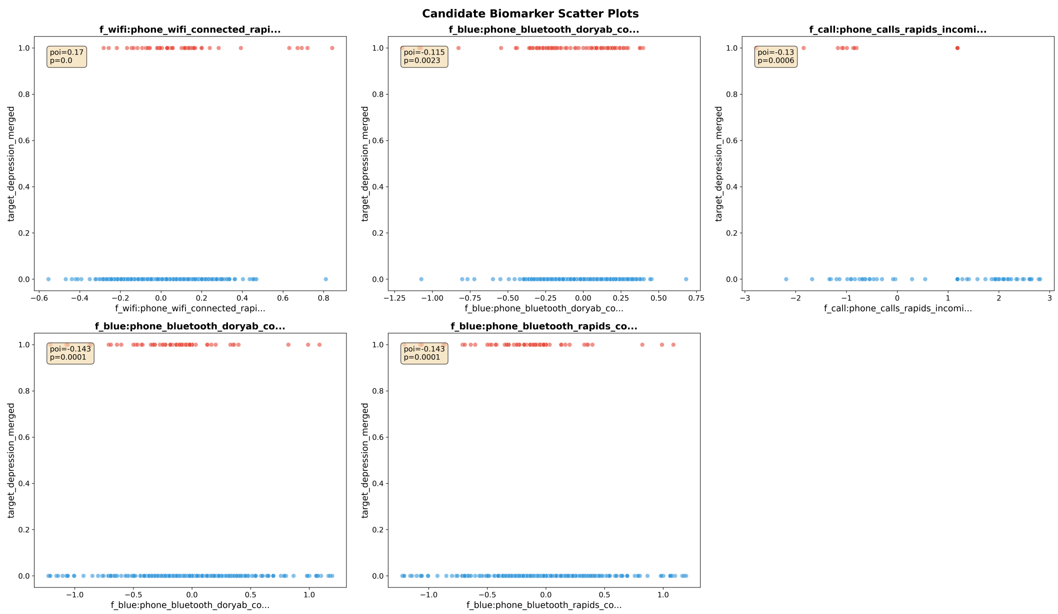

Figure 5: Scatter plots of validated biomarker candidates against target_depression_merged with linear regression lines. Points colored by above/below median target_depression_merged. Annotations show point-biserial correlation and p-values.

replication cohorts. Furthermore, given the severe class imbalance (87.8% non-depressed vs. 12.2% depressed), AUC alone may overstate discriminative ability; area under the precision-recall curve (AUPRC) should be considered as a complementary metric in future evaluations.

Missing Gender Information Gender distribution was not reported in the GLOBEM dataset. For a depression study, this is a significant limitation: gender is among the strongest known confounders for depression prevalence (approximately 2:1 female-to-male ratio in general populations), symptom presentation, and behavioral patterns. The absence of gender data precludes gender-stratified analysis, and the discovered biomarker associations may not generalize equally across genders. Datasets with gender metadata should be prioritized in future replication efforts.

Generalizability and Clinical Translation Furthermore, subgroup analyses for some conditional biomarkers were limited by "too small or constant" target groups within specific subgroups, potentially impacting the generalizability of these particular findings across all participant strata.

Finally, the discovery process revealed a Jaccard similarity of 1.0 between iterative rounds and a cross-method agreement of 0.0, with zero new candidates identified in later stages by alternative methods. This suggests that the pipeline rapidly converged on a very small set of statistical candidates without substantial new discoveries in subsequent rounds or significant agreement between different statistical and machine learning discovery methods. The absence of robust machine learning candidates identified during the initial discovery phase, coupled with the subsequently very poor

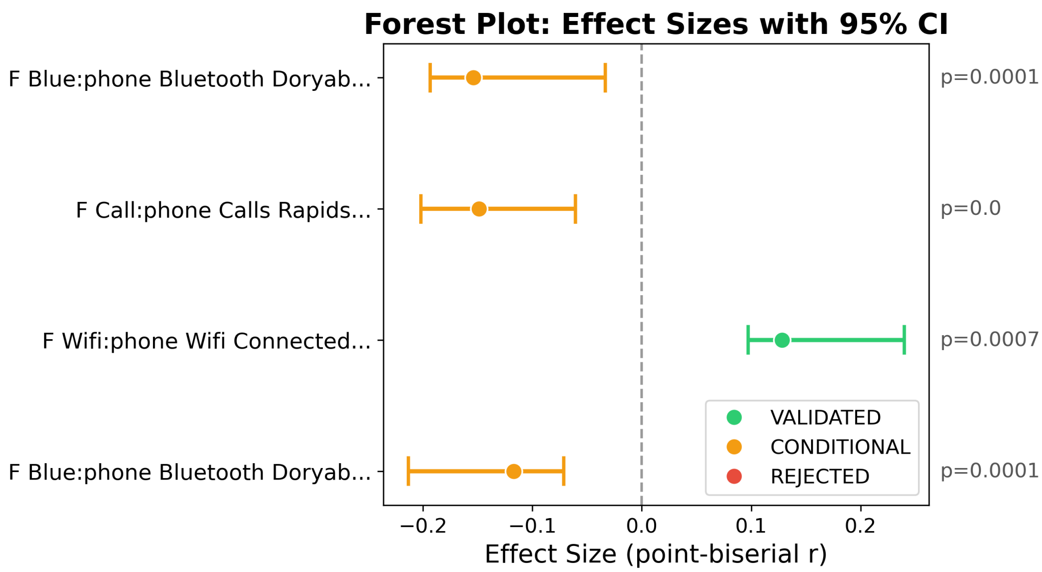

Figure 6: Forest plot of top biomarker candidates by effect size. Horizontal bars represent 95% bootstrap confidence intervals. Color indicates validation verdict: green (validated), orange (conditional), gray (insufficient evidence). Features are ordered by absolute effect magnitude.

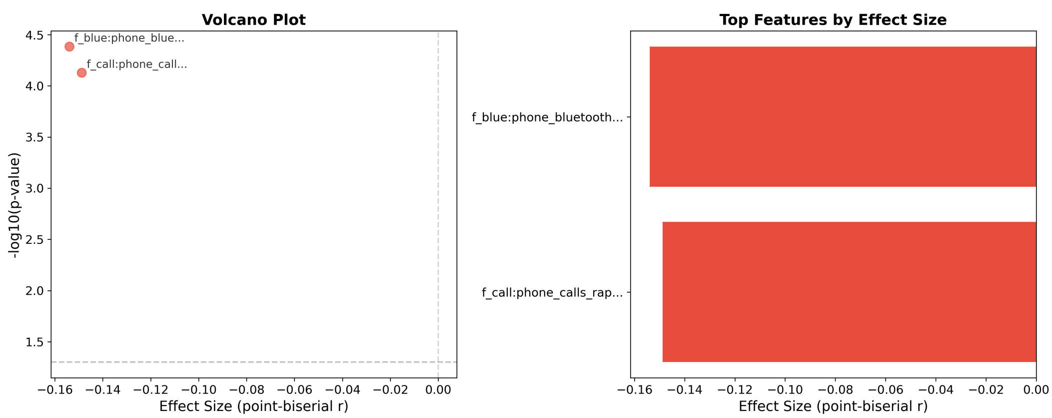

Figure 7: Statistical association results. Left: volcano plot showing effect size vs. statistical significance for all features tested. Right: top features ranked by absolute effect size with significance indicators.

performance of the predictive models on the full feature set, highlights a significant challenge in extracting robust predictive patterns for depression using current high-dimensional machine learning techniques on this specific dataset without careful feature curation.

Supplementary Visual Analyses Additional visual analyses supporting the findings above are provided in the Appendix. These include the subgroup activity level (Figure 8), the subgroup age groups (Figure 9), the subgroup analysis (Figure 10), and the target distribution (Figure 11). Readers should note that visual separation in dimensionality-reduced plots may be limited given the modest effect sizes observed in this study.

References

T. Bauer, K. Reutter, S. Schnieder, S. Eifler, P. Matusall, A. Ziemke, J. Meinhardt, J. Meyer, K. J. Bär, U. Hegerl, M. J. Müller, S. Strunz, and J. B. Zeller. Digital phenotyping for stress, anxiety, and mild depression: Systematic literature review. *JMIR Ment Health*, 11:e53549, 2024. doi: 10.2196/53549. URL https://menthealth.jmir.org/2024/1/e53549.

Yuxing Chen, Zihang Wu, Jingyi Wu, and Minghao Sun. Use of mobile sensing data for longitudinal monitoring and prediction of depression severity: Systematic review. *JMIR mHealth and uHealth*, 10(7):e34947, 2022. doi: 10.2196/34947. URL https://mhealth.jmir.org/2022/7/e34947.

S. Hossain, S. Akter, M. A. Rahman, and M. S. Rahman. Digital phenotyping and feature extraction on smartphone data for depression detection. In *Proceedings of the 2021 2nd International Conference on Robotics, Electrical and Signal Processing for Industrial Applications (RESPFIA)*, pages 235–240. IEEE, 2021. doi: 10.1109/RESPFIA54215.2021.9644342. URL https://ieeexplore.ieee.org/document/9644342.

R. Jain, T. Ma, J. Ma, S. Sanyal, X. Ma, C. Sun, Z. Han, and R. Lin. Digital biomarkers in depression: A systematic review and call for standardization and harmonization of feature engineering. *Journal of Affective Disorders*, 353:1–11, 2024. doi: 10.1016/j.jad.2024.03.048. URL https://pubmed.ncbi.nlm.nih.gov/38555230/.

D. Li, J. Huang, Y. Wang, L. Wang, Y. Wei, H. Zhang, L. Zhang, T. Zhu, and C. Lu. Detection of depressive symptoms in college students using multimodal passive sensing data and light gradient boosting machine: Longitudinal pilot study. *JMIR Mental Health*, 11:e46879, 2024. doi: 10.2196/46879. URL https://mental.jmir.org/2024/1/e46879.

David A. Rohani, Rubina Karunamuni, Harshini Thabrew, Jaesun A. Kim, Tapas Gangopadhyay, Michael E. Thase, and Thomas R. Insel. Digital biomarkers for depression screening with wearable devices: Cross-sectional study with machine learning modeling. *JMIR mHealth and uHealth*, 9(10):e31388, 2021. doi: 10.2196/31388. URL https://mhealth.jmir.org/2021/10/e31388/.

Yahya Saad, Reem Al-Ali, Aisha Al-Thani, Sara Al-Maadeed, Aisha Al-Khalaifa, Fatma Al-Mesaifri, Lulwa Al-Mulla, Salma Al-Saadi, Amani Al-Shafi, and Fatiyah Al-Thani. Improving depression severity prediction from passive sensing: Symptom-profiling approach. *Sensors (Basel)*, 23(21):8776, 2023. doi: 10.3390/s23218776. URL https://www.mdpi.com/1424-8220/23/21/8776.

J. Torous, J. Lorme, C. Corriveau, J. Huang, S. Han, K. Le, R. Yang, J. J. O'Connell, M. Milstein, J. P. Onnela, and P. F. Santangelo. Digital phenotyping of depression: A multi-modal passive sensing approach to identifying novel behavioral and physiological markers of treatment response. *npj Digital Medicine*, 4(1):1–10, 2021. doi: 10.1038/s41746-021-00438-w. URL https://www.nature.com/articles/s41746-021-00438-w.

A Supplementary Figures

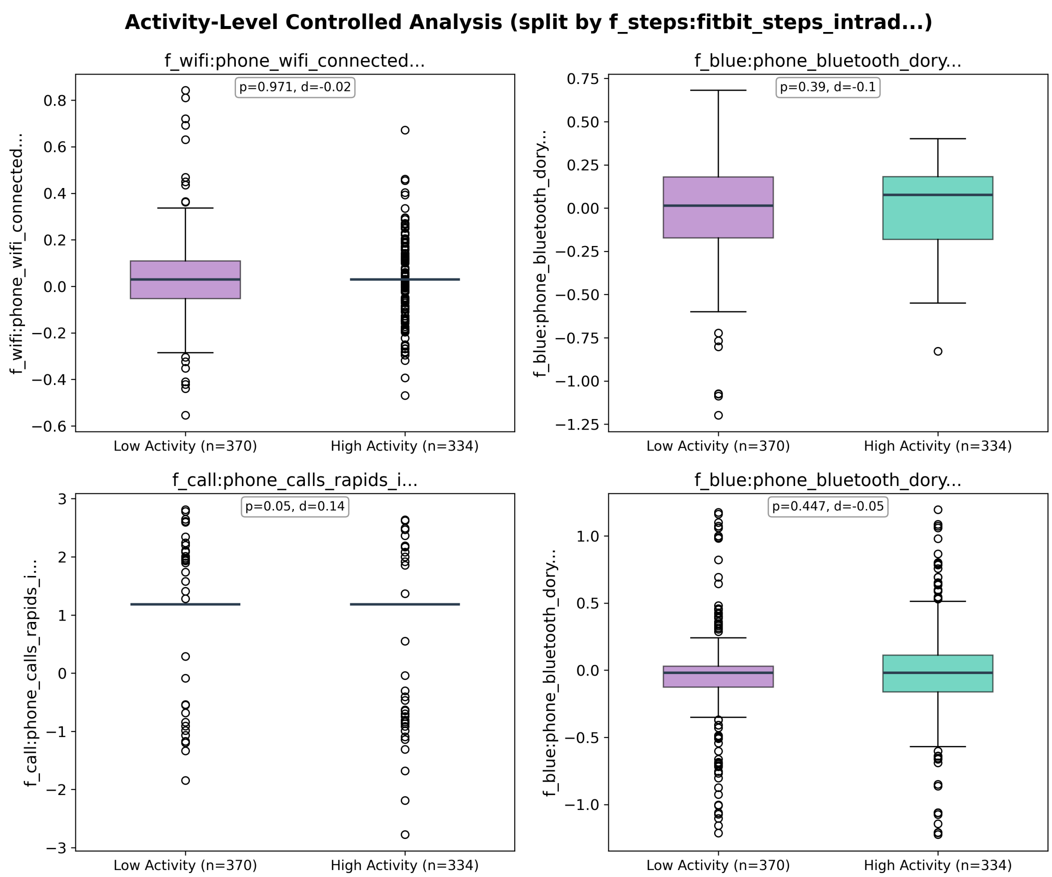

Figure 8: Activity-level controlled analysis for top validated biomarkers. Participants are split at the median physical activity level (e.g. daily step count) to examine whether the biomarker target_depression_merged relationship persists after controlling for physical activity.

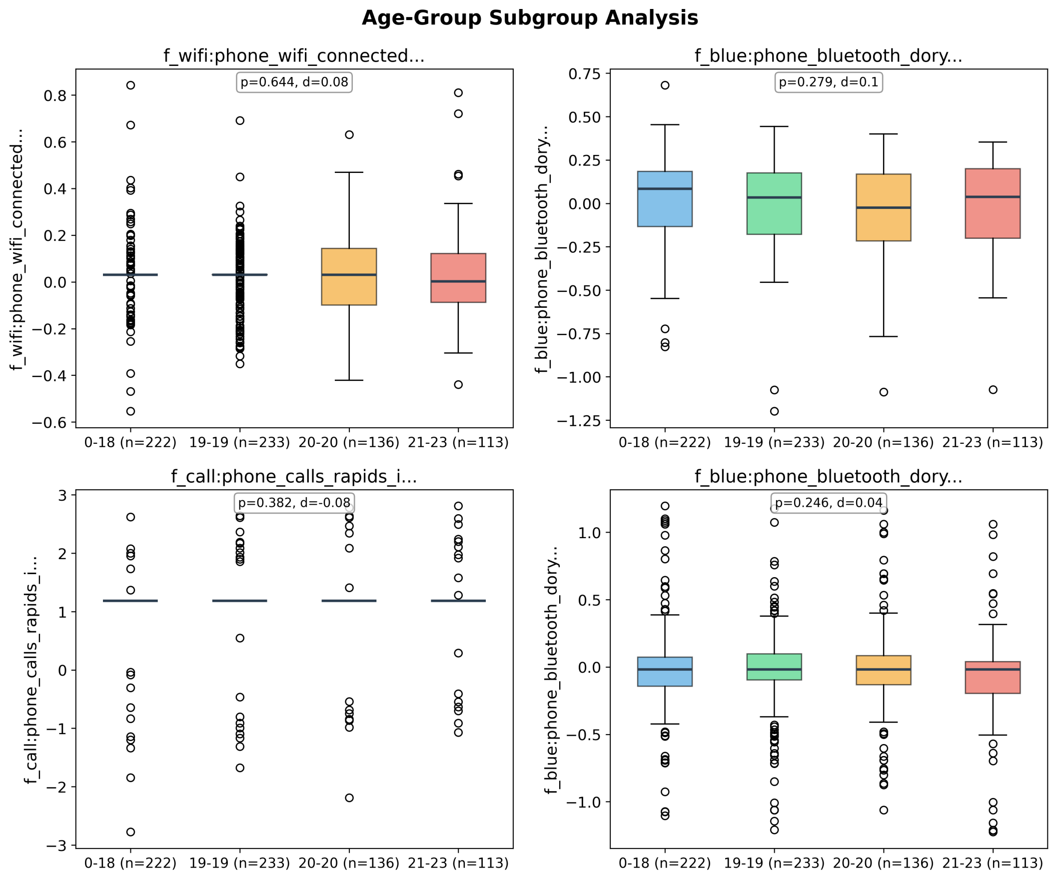

Figure 9: Age-group stratified subgroup analysis for top validated biomarkers. Given the study cohort's mean age of 19.2 (SD 1.4) with ages ranging up to 23, participants are binned into relevant age brackets to assess whether the biomarker relationship varies by age. Box plots show the distribution of each feature within each age group.

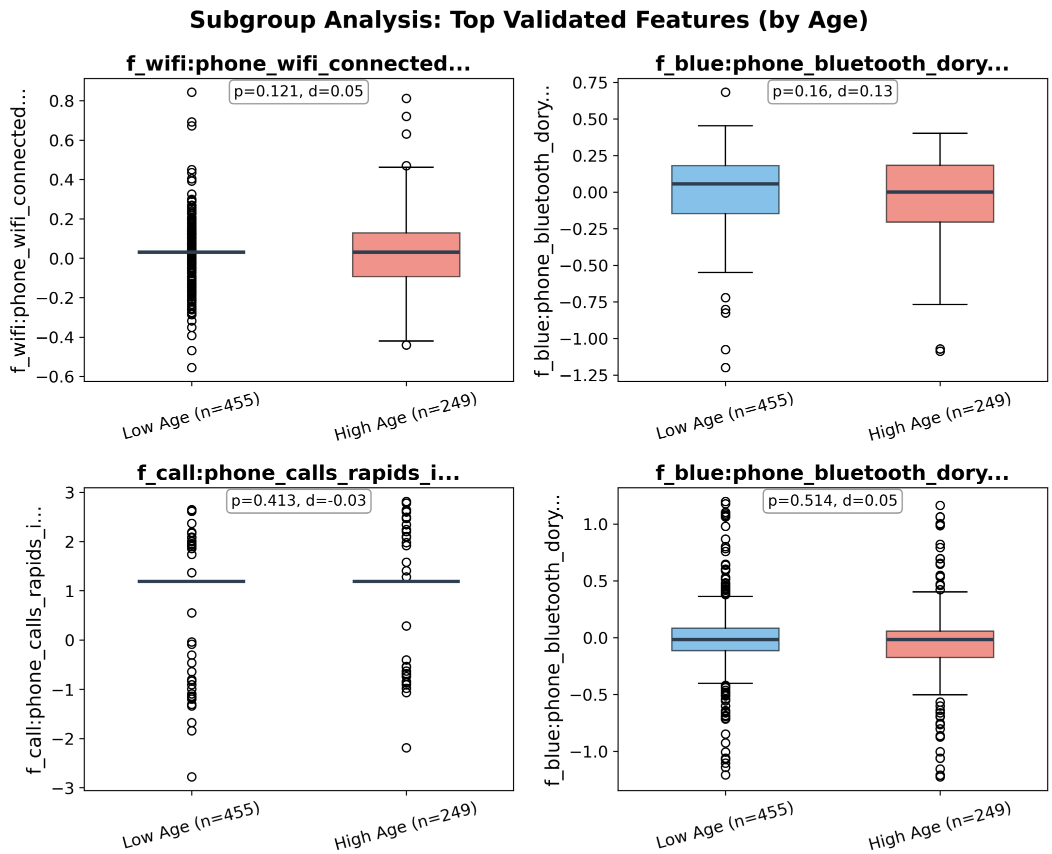

Figure 10: Subgroup analysis comparing feature distributions across participant groups for the top validated features. Box plots show group differences and distributional properties for key biomarker candidates.

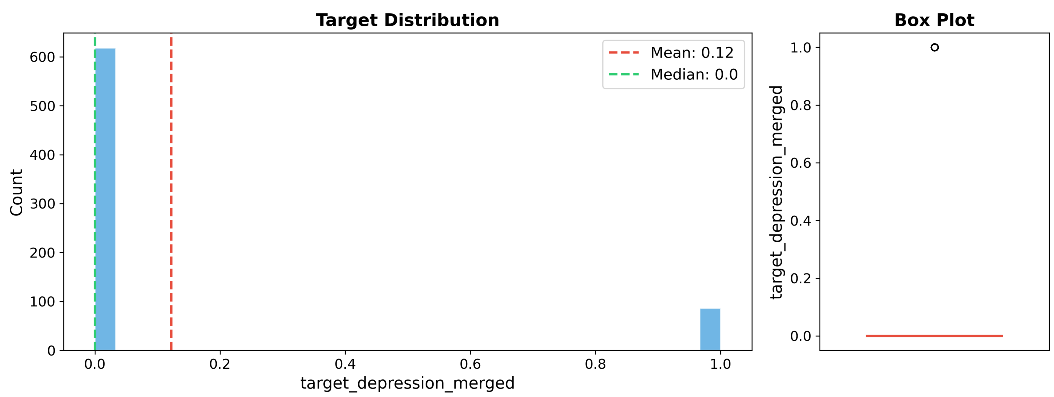

Figure 11: Distribution of `target_depression_merged` scores across the study cohort (N=704). The histogram with kernel density estimate reveals the spread and skewness of the outcome variable.

Table 11: Computational Cost Breakdown by Agent (Top 15)

Agent	LLM Calls	Input Tokens	Output Tokens	Est. Cost (USD)
literature_updater	5	468,962	16,334	$0.2122
gap_finder_agent	5	410,814	4,411	$0.2120
stat_interpreter	8	406,256	2,727	$0.2035
ml_interpreter	12	317,970	1,455	$0.1501
SearchExecutor	1	693,560	130,996	$0.1358
findings_reviewer_agent	1	257,107	1,602	$0.1310
LiteratureSynthesizer	1	211,770	3,342	$0.1109
PaperExtractor	1	1,036,610	20,810	$0.1099
SearchPlanner	1	173,079	1,066	$0.0881
strategic_assessor	1	164,968	313	$0.0830
scout_agent	1	147,747	1,375	$0.0759
mechanism_hypothesizer	1	84,187	3,833	$0.0478
novelty_classifier	1	81,894	1,338	$0.0430
hypothesis_generator	1	78,874	2,045	$0.0425
causal_model_generator	1	77,681	1,713	$0.0414
Total	**41**	**5,683,732**	**248,920**	**$2.31**

Analytical tasks (hypothesis generation, literature review, adversarial validation) used a larger, more capable model; text generation tasks (section writing, revision) used a faster model.